



RIBA PART 3  
EXAMINATION IN PROFESSIONAL PRACTICE

# **Spring 2021 Practice Problems**

## **Day 1**

### **Questions 1-5**

## **GUIDANCE NOTES**

In order to be registered as an architect under the Architects Act 1996 and to be eligible for chartered membership of the Royal Institute of British Architects, candidates have to be able to demonstrate their ability to deal competently with situations that arise in the ordinary course of general architectural practice. The practice problems of the RIBA Part 3 Examination in Professional Practice are devised in order to provide candidates with the opportunity to demonstrate their professional competence, working in realistic conditions over two days in their office.

Candidates are to imagine themselves as an employee of the architectural practice described in the scenario, taking the role of the Candidate, responding to each of the problems set out for each of the two days, using the information given in the scenario and in the particular problems. All of the questions carry equal weight and each of them must be attempted. The questions are issued at 0900hrs to the candidates each day.

Some of the questions may well require more investigation than others and the responses to the problems may well vary in length and complexity depending on the nature and circumstances involved. It is suggested that the Candidate first read through all of the questions before allocating an appropriate amount of time to each of the problems and allowing sufficient time to read through their responses before emailing them to their Examination Supervisor at 1730 of each day. The candidate should also upload a copy to the BOX.com links (provided via email on 24 February 2021) on each of the respective days.

Normally the total amount of material submitted in response to the problems is not expected to exceed 5,000 words. Where reference is made to published material the source and the page number must be cited; the copying of published material is not required and should not be necessary, information gained from websites must also be cited. Clear succinct responses are preferred to over long verbose discursive answers. Candidates are advised to ensure that their answers are in the form requested, for example where a question asks for a letter it is essential that the response is in the form of a conventional business letter.

Responses to the problems must be typed in A4 size format.

The information given in the scenario and in each of the questions is designed to be helpful and relevant; it is strongly recommended that it should be read carefully before making responses to the problems. Any assumptions made in the course of responding to a problem must be indicated and candidates must be prepared to discuss them during the professional interview.

Candidates may use any reference material, standard documents, the office library and the normal office aids including computers, photo-copiers, stationery, standard documents within the office, subject to the agreement of your employer. The practice problems and the responses to them must not be discussed with others until after their submission which must include a declaration notice signed by the candidate certifying that it is the candidate's own unaided work.

Where there is more than one candidate taking the examination within the same office, reference material may be shared but candidates may not discuss the problems or their responses to them until after their submission.

At the end of each of the two days at 1730hrs (or no later than 1930hrs if the candidate has been granted an additional 2 hours), candidates are required to email their Examination Supervisor with copies of their draft responses which should be retained by the exam supervisor. Candidates are also required to upload a copy to the unique BOX.com links (provided via email on Wednesday 24 February 2021) on each of the respective days.

1730hrs Wednesday 03 March 2021 – Day One Practice problem responses

1730hrs Thursday 04 March 2021 – Day Two Practice problem responses

*NB – responses for each exam day should be saved in PDF format. The filename should read as the following: 'Practice Problem Responses Day One – Candidate Name' & 'Practice Problem Responses Day Two – Candidate Name'.*

*e.g. 'Practice Problem Responses Day One – Joe Bloggs'*

The uploaded responses are devised as a means of protecting the integrity of the examination; they will be used by the examiners in comparing them with the final versions in the documentary submission. Any deviation between the draft and final answers other than the allowed proof reading and typographical correction, formatting and presentation can result in the Practice Problems being failed for not complying with the regulations.

The third day, Friday 05 March 2021, is solely for the copy typing, proof reading and typographical correction, collation and spiral binding of two copies of the documentary submission which have to be dispatched together with the Declaration certifying that the whole submission is the candidates own unaided work.

The cover of the documentary submission must show the candidate's full name, the title of the practice in which the responses to the problems were written and the name of the Examination Supervisor. Candidates are advised to prepare a copy of their submission for reference in preparation of the professional interview and if need be in the course of the interview.

The parcel containing these documents must be securely packed and carefully addressed to Alison Mackinder, Professional Education Programmes Director, RIBA Part 3, RIBA North, 21 Mann Island, Liverpool Waterfront, **L3 1BP**

Parcels must be dispatched by 1730hrs on the third day using the Post Office Guaranteed Delivery System or its equivalent and a timed receipt obtained for submission to the examiners at the professional interview.

Alternatively, if using the Hobs print, bind & delivery service, you must upload your submission to their portal by 1730hrs on the third day, using the guidelines previously provided.

You also need to upload your submission prior to 1730hrs to the unique BOX.com link (provided via email on Wednesday 24 February 2021).

**Final Submission: 1730hrs on Friday 05 March 2021**

## SCENARIO

The name NDG Architects Ltd is listed in the current RIBA Directory of UK Chartered Practices.

Established in 1998, the practice has a wide range of design and development experience across a number of private-sector and public-sector areas, including commercial, retail, industrial, education and health care.

Exclusively based in the North West of England, the practice office is located in a converted mill building in Central Madchester. As confidence in the workload grows, the practice is looking at opportunities to expand and develop its portfolio. A number of changes were introduced to the structure and management of the office in 2020 with the retirement of founding partner, James Noble, the appointment of Sue Allen as a Director and the promotion of Sven Berg to Associate Director and Renata Wang to Senior Architect. There have been some changes to personnel in the last six months, as two year-out students have returned to start their Part 2 studies, Olga Poulicek has successfully completed her Part 3 examination and is now enjoying life as an architect within the practice whilst new Part 1 and Part 2 graduates have joined the practice as the office continues to deliver projects and market fresh initiatives to maintain workload in what have been very challenging times!

As a candidate for the Spring 2021 examination, you are required to assume that you are a salaried employee of NDG Architects and have been engaged at this practice for a period of 20 months whilst preparing to sit the Part 3 examination. Companies House records now show that NDG Architects Ltd has three directors and one associate director.

**Belinda Carter** is an architect and the Managing Director. Her role in marketing the practice continues with significant success this year, as a result she is leading a further review to determine the potential for the practice to restructure and expand into new project sectors. As part of this review she is eager to consider the views of all members of staff.

**Michael Hart** is a conservation architect and Director. With the retirement of James Noble, Michael has become a director of the practice. He is a RIBA specialist conservation architect and listed on the Diocese schedule of approved architects. His reputation in this field has previously resulted in the practice being commissioned to carry out larger conservation projects attracting lottery funding. Michael is also listed as a Client Design Advisor on the RIBA's CDA register.

**Sue Allen** is an architect and Director. She is an accredited sustainability expert and has completed a MSc course in Advanced Sustainability at the University of Madchester. Sue also acts as the company's quality manager, responsible for the ongoing development and maintenance of the quality assurance system and manual.

**Sven Berg** is an architect and Associate Director. He has exceptional technical skills and has taken on the role of Principal Designer within the practice after completing several courses aimed at designers wanting to provide this service and to explore the practicalities of design risk management.

Current practice members, roles, salaries and charge out rates are indicated in the table below:

| Staff and salaries/director remuneration |                                 |                    |                 |
|------------------------------------------|---------------------------------|--------------------|-----------------|
| Charge-out Hourly Rates                  |                                 |                    |                 |
| Name                                     | Role/Status                     | Salary             | Charge-out Rate |
|                                          |                                 |                    |                 |
| Belinda C Carter                         | Director                        | £ 60,000.00        | £90/hr          |
| Michael Hart                             | Director                        | £ 60,000.00        | £90/hr          |
| Sue Allen                                | Director                        | £ 60,000.00        | £90/hr          |
| Sven Berg                                | Associate Director              | £ 48,000.00        | £75/hr          |
| Dareem Patel                             | Senior Architect                | £ 40,000.00        | £60/hr          |
| Renata Wang                              | Senior Architect                | £ 40,000.00        | £60/hr          |
| Tomas Hjeltne                            | Architect                       | £ 35,000.00        | £55/hr          |
| Koari Furuta                             | Architect                       | £ 35,000.00        | £55/hr          |
| Timothy Wells                            | Architect                       | £ 35,000.00        | £55/hr          |
| Olga Poulicek                            | Architect                       | £ 35,000.00        | £55/hr          |
| George Gee                               | Architectural Technologist      | £ 30,000.00        | £45/hr          |
| Candidate                                | <b>Architectural Assistant</b>  | <b>£ 27,000.00</b> | <b>£40/hr</b>   |
| Kai Xin Lee                              | <b>Architectural Assistant</b>  | <b>£ 27,000.00</b> | <b>£40/hr</b>   |
| Mark Jones                               | Architectural Technician        | £ 24,500.00        | £36/hr          |
| Jamie Wang                               | Year-out Student                | £ 20,000.00        | £28/hr          |
| Sarah Jane Key                           | Administrative Assistant        |                    |                 |
| Sophie Bright                            | Receptionist/Clerical Assistant |                    |                 |
| Jane Dudley                              | Clerical Assistant              |                    |                 |
| Simon Shaw                               | Accounts Assistant              |                    |                 |

## **Covid-19**

The practice has continued to adapt to the restrictions imposed by the Covid-19 pandemic and whilst the office has been primarily closed since the lockdown was introduced, remote working has been very successful albeit not ideal. Unfortunately, a number of projects were put on hold and were subsequently cancelled or are still on hold with the respective clients needing greater confidence before committing to future investment. Happily, a number of projects that had been stopped at feasibility stage, have since re-commenced and are now moving forward. Several projects on site whilst initially delayed did recommence where the nature of the work despite the constraints imposed, allowed continuity. Several new projects introduced during 2020 are now also under construction. NDG has been reasonably fortunate in attracting new projects which has maintained a flow of work particularly at RIBA Work Stages 0-3 but cash-flow is tight. Whilst several members of staff were initially kept on furlough, the office is now back to full capacity, new staff introduced and controlled albeit fragmented use of the office by selective personnel as the constraints safely allow.

## **Significant projects currently being handled by the practice include:**

### **DZZ Heights Residential Development**

The practice was appointed as architects on an 18-storey dockside residential development which was completed in July 2016. The practice was appointed by the client to obtain planning consent and prepare tender documents and was subsequently novated to the design and build contractor who unfortunately went into administration last year. Most of the external envelope is constructed of aluminium cladding with lightweight insulated panels and was designed in accordance with the 2010 Building Regulations. The scheme was very successful, and all apartments sold quickly due to the waterfront location. The client has since sold the freehold to the residents' management company and no longer has any involvement in the site. We have also given the client a collateral warranty which has not been assigned to the new owner even though there is a clause which allows assignment.

### **New Sports Centre at Strange Hill School, Madchester**

NDG has maintained involvement in developments at Strange Hill School which has seen notable success at local and regional level in the field of athletics. A generous gift provided by a former pupil, Cheryl Bolt, who now runs a renowned chain of sports retail stores will provide specialist all-weather training facilities to heighten the athletic achievements of the school. A JCT SBC16 with quantities was selected for use and despite delays to programme due to Covid-19, all eyes are focused on September 2021 which will be the official launch of the £3.5m facility to be opened by a member of the royal family. One is very excited!

The programme is tight and the envelope is due for completion by Easter 2021 to allow application of specialist internal finishes and fit-out in the third quarter of 2021 in readiness for the grand opening.

### **Acorn Centre, Madchester**

NDG was approached by a local Planning Consultant to provide advice on a proposed re-development of a listed building within a deprived area of Madchester for the *Acorn Centre*, an adult education provider that has some ambitious plans for expansion of its current courses and greater outreach to the local community.

The Grade II building originally a detached Georgian house that was subsequently attached to the end of a Victorian terraced housing development, was formerly owned by a local developer but having suffered neglect, vandalism and a recent fire, resulted in the Local Authority obtaining the property through compulsory purchase (CPO). As part of a strategy for re-use required by the CPO, the Local Authority offered the building to the Acorn Centre to expand the existing and abutting accommodation within the converted Victorian terrace. Subsequent concept proposals prepared by NDG following appointment as architects, have been well received by the Local Planning Authority and the local community following pre-application consultation and the client, having secured funding, now wants to move quickly forward.

### **New House for Alec Smart**

The practice has been commissioned for RIBA Work Stages 1 – 7 for a new build house on a steeply sloping site and is appointed under a RIBA Standard PSC 2020 for architectural services with a separate appointment for Principal Designer. The project will be traditionally procured and the practice will act as Contract Administrator. NDG is providing Architect, Lead Designer, Project Manager and Principal Designer services. The Design Team includes QS, Civil/Structural Engineer and Building Services Engineer. The project is due for completion in December 2021.

### **Refurbishment of Tappers Dance Studio, Madchester**

The practice has recently completed the refurbishment of Tappers Dance Studios in Madchester. The £1m project was funded by a generous grant from Madchester Council and the owners of the studio were eager to see development finish as it now provides up-to-date facilities and is attracting dancers with wider interests. Whilst the project, carried out under a traditional procurement route, hit some initial budget problems and subsequent delays, the project reached successful conclusion and the client was delighted with the result albeit Covid-19 has limited capacity of the building and the full potential of the facilities is yet to be realised. The contract is in the Rectification Period with no significant defects reported and the feedback in use is very positive.



***NB:*** All of the characters, locations and projects of the scenario and the practice problems are entirely fictitious and any similarity or resemblance to persons, places, projects or practices is totally coincidental.

***NB:*** The dates of the emails to the candidate which outline the questions will not necessarily be sequential - for instance where the same project forms the basis of two separate questions which are set at different points/work stages.

**END**

## QUESTION 1

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**NDG Architects**

### INTEROFFICE MEMO

Date: 22 February 2021  
To: All staff members  
From: Belinda Carter  
Re: **Beyond Covid-19 and the lessons learned**

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Good morning All,

With the prospect of the end of the pandemic some way off, I think we should reflect on the last year and consider what we have learnt from changing our working practices. Whilst I cannot imagine a time when we will all be working remotely on a permanent basis. I think there are certainly some lessons that we can learn from a very challenging year.

I propose that we set aside a day in the next couple of weeks when we can all share our thoughts. In the meantime, I have put together an outline of an agenda to give some structure to our discussion and I would be glad if you could give me some initial thoughts on each of the areas based on your experience in 2020.

1. What have been the particular challenges in working remotely and how could we, as a practice, support the staff better?
2. How have we maintained effective communication between ourselves, with our clients, consultants and contractors. How could this be improved?
3. It will undoubtedly be a difficult period in the coming months with many of the projects delayed or on hold for the foreseeable future, the Government job retention scheme and loans are helping but we need to review our client profile and potential sectors that we can target in the future together with any further services that we can offer – any thoughts are welcome.
4. We are conscious that CPD has largely been ignored whilst we have been grappling with the current situation and we would be grateful of your thoughts as to how we can meet our professional obligations in 2021.
5. What lessons have we learnt from the past year and what positive measures can we take forward and develop for the future?

I look forward to hearing your views and thoughts.....Belinda

## QUESTION 2

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*Sven.Berg@ndgarchitects.com*

**EMAIL**

**NDG Architects**

**Date** 22 February 2021  
**To:** Candidate@ndgarchitects.com  
**From:** Sven Berg  
**Re:** **DZZ Heights Residential Development, Lynton: External Cladding**

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Hi,

Sorry for the late email, but DZZ Heights, the dockside residential building that was completed in 2016, has a new management company/owner who have contacted me requesting some information on the external envelope of the building. The request appears to be reasonable and they are offering to pay for our time. There is also a form which they would like us to complete.

I am under pressure to get the pre-construction information package together for the T6 Office Development due out to tender on Friday, so your help would be appreciated. Can you let me have you thoughts on this matter, in particular:

- a. Is there any reason why we cannot sign the form?
- b. Is there any reason why we should not give them pdf drawings for their records?

Could you also please draft a letter to the management company with our response to their letter for me to check and discuss with Belinda before sending.

Thanks and best regards

Sven

*Sven Berg*  
**Associate Director**

## QUESTION 2

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Sven Berg  
*NDG Architects*  
Old Mill Building  
Maccchester  
MX1 9DS

Dear Mr Berg,

### **DZZ Heights, Residential Development, Lynton**

I understand you were the project architect for the above development. I write on behalf of the Dockside Residents Committee and would firstly like to thank you for designing such a lovely building. The residents and visitors all compliment your design and love the large spacious rooms with floor to ceiling windows overlooking the old docks.

We would be very grateful for your assistance. Several apartments have been resold recently and some of the new purchaser are having some difficulty in obtaining mortgages as there is insulated cladding on the building. We know the build is 'pre-Grenfell' and designed to old regulations, but the banks are asking if the attached EWS1 form could be completed by the architect and returned to them. I assume you will have the knowledge and information to be able to assist with this.

I have also received a set of 'as-built' drawings from the previous owner, but they are only plans and elevations and do not give any construction information. I wonder if you could also send me a full set of your drawings or specifications indicating what the contractor has built. This will be useful for our records and will also help us with any future maintenance. I acknowledge there will be a cost for your time spent on this and will gladly reimburse you for your time.

I look forward to hearing from you.  
Yours faithfully

Simon Piper  
*Director*  
*Dockside Developments Limited*

DDL Tower  
Sea View Annexe  
Floor 9  
Flowmouth  
FL2 5RY

COPY OF FORM RECEIVED FROM DOCKSIDE DEVELOPMENTS

| Subject Address (One form per block) |        |      |                       |
|--------------------------------------|--------|------|-----------------------|
| Block or building name               | Street | Town | Postcodes (all built) |
|                                      |        |      |                       |

I confirm that I have used reasonable skill and care to investigate <sup>(Note 5)</sup> the primary external wall materials (typically insulation, filler materials and cladding) and attachments of the external walls of the above building/block.

**OPTION A<sup>(Note 1)</sup> – Where external wall materials are unlikely to support combustion**

I confirm that:

- I meet the professional body membership and competence criteria as described in Note 2
- In relation to the construction of the external walls, to the best of my knowledge the primary materials used meet the criteria of limited combustibility <sup>(Note 6)</sup> or better and cavity barriers are installed to an appropriate standard in relevant locations (Note 7)
- In relation to attachments to the external wall (*tick one of the following*):
  - ☐ A1 - There are no attachments whose construction includes significant quantities of combustible materials (i.e. materials that are not of limited combustibility <sup>(Note 6)</sup> or better);
  - ☐ A2 - There is an appropriate risk assessment of the attachments confirming that no remedial works are required
  - ☐ A3 – Where neither of the above two options apply, there may be potential costs of remedial works to attachments <sup>(Note 8)</sup>

**OPTION B<sup>(Note 1)</sup> – Where combustible materials are present in external wall**

I confirm that:

- I meet the professional body membership and competence criteria as described in Note 3
- I have used the reasonable skill and care that would be expected of the relevant professional advisor to assess the level of fire risk <sup>(Note 9)</sup> presented by the external wall construction and attachments (*tick one of the following*)
  - ☐ B1 - I have concluded that in my view the fire risk <sup>(Note 8)</sup> is sufficiently low that no remedial works are required
  - ☐ B2 - I have concluded that an adequate standard of safety is not achieved, and I have identified to the client organisation the remedial and interim measures required (documented separately).

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|              |       |                   |       |
|--------------|-------|-------------------|-------|
| Name         | ..... | Qualifications    | ..... |
| Organisation | ..... | Professional body | ..... |
| Signature    | ..... | Date              | ..... |

### QUESTION 3

Page 1 of 3

*Sue.Allen@ndgarchitects.com*

**EMAIL**

Date: 23 February 2021  
To: Candidate  
From: Sue Allen  
Re: **New House for Alec Smart**

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As you know we are nearing the end of RIBA Work Stage 4 and the client, Alec Smart, is unhappy with the QS estimate for the new retaining wall which forms an integral part of the lower floor accommodation on this steeply sloping site.

This is a traditionally procured project and we have begun to put together a list of potential general contractors for consideration by the Client. However, Alec Smart intends to take the retaining wall out of the contract (shown in red on the attached section) and get it built before the main contract starts using a specialist firm of contractors he knows. We have expressed concern about this approach, but the client is determined to do this in order to save money and says he will manage this part of the project himself. The specialist contractor will both design and construct this element of the new house and all their work will be completed before the Principal Contractor starts the main contract works.

I do not want you to concern yourself with the contractual ramifications (NDG will not be involved at all), but I would be grateful if you could consider the situation from a Health and Safety point of view and prepare a Design Risk Assessment (DRA), just for the retaining wall, using the attached format identifying 'Red' 'Orange' and Green' risks. I have attached a Word Document version of the template to this email for your use. Identify just four principal items!

I also want you to prepare a memo addressed to me explaining the following:

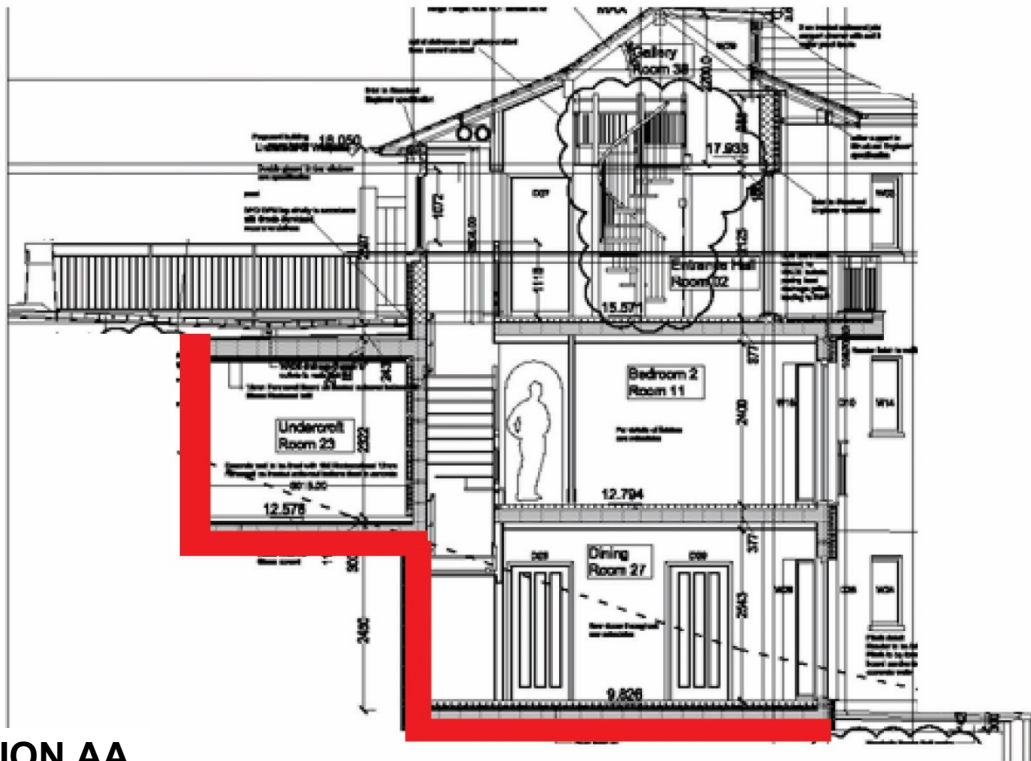
- The status of the DRA at the point we hand it to the client's specialist civil engineering D+B contractor (who, for this element will also be the Principal Contractor)
- Who else we might involve in putting together the initial DRA?
- What steps we need to take after the client appoints the contractor, but before they start on site and who else we need to involve?
- Where the client fits in, given NDG is Principal Designer but not Contract Administrator?
- What needs to happen at the completion of the retaining wall?
- Can you see any other safety or regulatory issues associated with all this?

Thanks.....Sue

### QUESTION 3

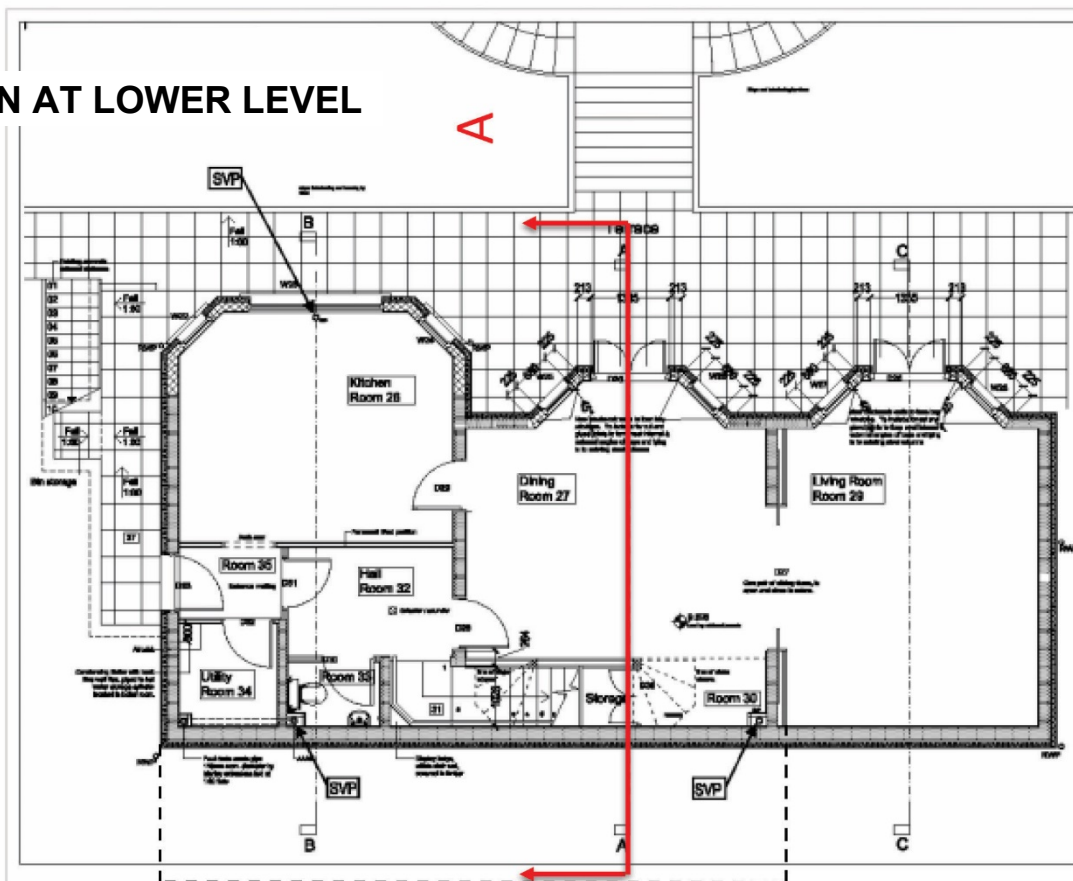
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#### Annotated Drawings: House for Alec Smart



SECTION AA

PLAN AT LOWER LEVEL



**DRA Template (Word Document Version also attached)**

[illegible]



## QUESTION 4

Page 1 of 2

**NDG Architects**

### INTEROFFICE MEMO

Date: 24 February 2021

To: Candidate

From: Belinda Carter

Re: **A tough year!**

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It has been a very tough twelve months!! Our cash-flow is always tight in January/February and just as things were starting to pick up, Covid-19 arrived. We suddenly had very little work and drastically reduced income. We furloughed most of our staff, but the bills still arrived. We paid thousands of pounds towards rent, rates, insurance, phones, leases, etc for an empty office we did not use for over 6 months. We got a grant payment from the government which paid some bills, and the bank gave us a cheap loan which we do not have to repay until next year

Fortunately, we are all back working, using the office in flexible and varying degrees and things look a lot better, but it will be some time before we are back to normal. We are in a very difficult cash-flow position as we do not have sufficient funds to pay all our bills but have more than enough work on the horizon and will need more staff if all our initiatives bear fruit. Last year was profitable and next year could be very profitable but unless we do something quickly, we may have to downsize very soon and possibly stop trading within the next two months if we are pushed by our creditors.

We have a couple of regular/loyal clients who consistently give us work and always pay quickly. This is not very profitable work, but it is reliable and it is how we usually survive from month to month. We have a large residential project starting on site soon with a new design and build contractor which will give us 24 months income. The contractor will need a lot of information from us in the first 3 or 4 months, but we do not have the resources to serve the project and do not have the funds to employ more staff at the moment.

We are also shortlisted in a competition for a community project and must quickly produce a design for the final stage. This is a very prestigious project, and we have a good chance of winning, but we must find the resources to do the work. There are also a couple of good projects which have been on hold for the last year and the clients have contacted me recently to discuss moving these projects forward.

**QUESTION 4**  
**Page 2 of 2**

**NDG Architects**

**MEMO**

Date: 24 February 2021

Re: **A tough year!**

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**CONTINUATION SHEET**

Only a few people in the practice, selected across the various layers of experience, have been made aware of the situation in order to canvas opinion. I realise you have limited experience, but you have freshness and freedom of thought, so I would welcome your views on how we could manage this problem and I would, ideally, like you to suggest some strategic ideas to get us through the next six challenging months.

I would also value your opinion on whether profit or cash-flow should be our long-term priority.

Please keep the contents of the memo confidential. I look forward to hearing from you shortly once you have had a chance to digest the situation.

Thanks and best regards

Belinda

Belinda Carter

*Director*

**NDG Architects**

**QUESTION 5**  
**Page 1 of 3**

*Michael.Hart@ndgarchitects.com*

*EMAIL*

Date: 26 February 2021  
To: Candidate  
From: Michael Hart  
Re: **New Sports Centre at Strange Hill School, Madchester**

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Good afternoon,

Belinda asked me to take a look at the progress and quality of the work on site of the new sports centre at Strange Hill School in North Madchester. The project followed the successful refurbishment work of the gymnasium in the main school building and a generous gift provided by a former pupil, Cheryl Bolt, who now runs a renowned chain of sports retail stores and wants to provide specialist all-weather training facilities to heighten the athletic achievements of the school. The benefactor was insistent that the contract is carried out under a traditional procurement route as she has had bitter experience of D&B contracts in her last two store projects. A JCT SBC16 with quantities was selected for use. Everything is focused on Monday 13 September 2021 which will be the official launch of the £3.5m facility that is being opened by a member of the royal family. The programme is tight and the envelope is due for completion by end June 2021 to allow application of specialist internal finishes and fit-out during the final two months of the contract period.

The brickwork started several weeks ago but I had not managed to arrange a visit until yesterday. The contractor had made good progress and the brickwork was almost at first floor level. However, upon closer inspection, I considered the quality of work poor, so I have asked the contractor to remove the facing brickwork down to the engineering brick plinth level and re-construct the walls. There is a valuation due at the end of this week, so I have already advised the cost consultant to omit any costs associated with the facing brickwork.

I am unhappy with overall appearance. There is significant variation in the brick, well outside the tolerances set by the British Standards and also highlighted in the specification. I think the contractor has not mixed the packs well enough on site; the perpends do not line through vertically; there are odd-shaped bricks of differing sizes which should not have been used; the recess joints are messy and there is variation in the mortar colour.

The Head Teacher is worried about the programme and the immovable opening date, so he has emailed me this morning following my visit yesterday which now needs an urgent reply.

**QUESTION 5**  
**Page 2 of 3**

*EMAIL*

*CONTINUATION SHEET*

Date: 26 February 2021  
From: Michael Hart  
Re: **New Sports Centre at Strange Hill School, Madchester**

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The Contractor has also left me a voicemail message requesting a formal Architect's Instruction to remove and rebuild the brick walls which I am reluctant to issue as it is not required. Under the terms of the SBC/Q 2016, we are entitled to request removal of work that is not to our satisfaction.

There was a lot of thought put into the selection of the brick mindful of the location and setting on the edge of the playing fields and external running track with the dense woodland as backcloth. The specification clearly cross references to workmanship and the relevant British and European standards but I am starting to wonder whether I have been a little hasty to react.

Can you please email your thoughts on the matter and draft an email in reply so I can quickly review and formally respond when I am back in the office later this afternoon.

Thank you and best regards

Michael

***Michael Hart, Director***  
***NDG Architects***

**QUESTION 5**  
**Page 3 of 3**

*EMAIL*

Date: 25 February 2021  
To: Michael Hart  
From: Timothy East  
Subject: *New Sports Centre at Strange Hill School, Madchester*



*Strange Hill School  
Athletics Academy*

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Dear Mr Hart,

I realise Belinda is away at the moment and you are acting on behalf of NDG. I am writing regarding your verbal instruction to the contractor yesterday to remove quite a substantial portion of constructed brickwork.

I respect and support your judgement as our contract administrator, but I am extremely concerned that your instruction will cause a delay and we must be finished, fully operational and ready for the VIP opening in September 2021.

Personally, I do not have an issue with the appearance of the bricks as NDG Architects persuaded the Steering Group to have a rustic finish but now that you have highlighted the shortcomings of the contractor, I agree that it does not look right. However, the contractor has also contacted me and feels that the bricks and mortar have been used in accordance with your specification.

The Steering Group is not experienced in construction so will have to rely on the advice offered by our appointed professionals. We would however like some reassurance that every effort will be made by the contractor to finish the building on time and that we will not incur any additional cost as a result of this matter.

Can you please respond as quickly as possible.

Yours sincerely

Timothy East  
Head Teacher

One final thought: I don't remember seeing any brickwork samples on site, is this usual?

**END OF DAY ONE**

**Please ensure that you email a copy of your draft answers  
to your exam supervisor  
at 1730hrs**

**&**

**Please ensure that you upload a copy of your draft answers  
to the box.com link (provided in the exam scenario email sent 24 February 2021)  
at 1730hrs**



RIBA PART 3  
EXAMINATION IN PROFESSIONAL PRACTICE

## **Spring 2021 Practice Problems**

### **Day 2**

### **Questions 6-10**

## QUESTION 6

Page 1 of 2

[jack.lumber@homeofficesolutions.co.uk](mailto:jack.lumber@homeofficesolutions.co.uk)

EMAIL

**Date** 01 March 2021  
**To:** candidate@ndgarchitects.com  
**From:** Jack Lumber  
**Re:** Home Office Solutions



It was a remarkable coincidence bumping into you whilst jogging around Madchester Park (figuratively speaking of course, within these social distancing times) and I had no idea that you were now working for NDG Architects. It was good to share with you albeit briefly, the exciting ideas that my sister Theresa and I have been developing since taking over my father's joinery company that has struggled to keep a steady flow of work and orders since the pandemic struck.

We hit on the idea of rapid and easy assembly, pre-fabricated detached buildings for office use in these times of remote working, inspired by the gazebo we built for mum and dad in the garden, the availability of a fully equipped large joinery workshop and the outpouring of interest since we aired the idea to friends, acquaintances, work colleagues and the bank!

We feel that there is clearly a market for these simple, cost effective and flexible eco-buildings that can provide additional space without impact on the main house for use as an office in the current climate but convertible to a garden room for a multitude of potential uses in the future. However, we recognise our design skills are limited and we think that we may be missing an opportunity for wider markets as enquiries seem to be gathering momentum with more demand for a complete package.

We have been reluctant to advertise for an architect as we need someone talented but also someone we can work with and will fit-in - hence this follow-up email. I realise that you will not want to leave NDG at the stage you are at with imminent completion of your Part 3, but I wondered whether there was possibility of an association with NDG or with you as an individual that would allow us to market a design and build a package to the mutual benefit of us all?

Would you be willing to give the suggestion some thought and perhaps we can arrange a virtual meeting with you, me and Theresa to discuss the opportunities and explore some basic principles that will allow us to move forward. Time is pressing and interest is mounting so we would like to chat as soon as possible.



## QUESTION 6

Page 2 of 2

### EMAIL

**Date:** 01 March 2021  
**From:** Jack Lumber  
**Re:** Home Office Solutions

### CONTINUATION SHEET



I have generated a few questions which hopefully you can consider before we meet:

- a. Is it of interest to you?
- b. Would it be best pursued with you as an individual or with the practice?
- c. Would it be best as a separate company?
- d. What are the advantages and disadvantages?
- e. Is there any restriction to you as an architect (once you register after Part 3) being involved?

These are just a few of my own thoughts, so please feel free to expand on these aspects but let us keep the financial side out of it at the moment and simply look at the principles and practicalities of the idea.

Theresa and I really hope you will get involved!

Take care and best regards

Jack

*Jack Lumber*

**Director**

**Home Office Solutions**

**NDG Architects**

**Date** 01 March 2021  
**To:** Candidate@ndgarchitects.com  
**From:** Belinda Carter  
**Re:** **Union Street Madchester: Negligence Claim**

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Hi, I hope all is well.

On top of everything else at the moment, Jim Mills and ABC Holdings are suing NDG Architects for negligence regarding a feasibility study the practice undertook for Jim Mills in 2016.

The property in question was an office building in Union Street, Madchester, which Jim Mills was purchasing with a view to converting it into apartments. The sketch proposal indicated that it was possible to create sixteen apartments within the shell of the existing building. We attached a schedule of accommodation detailing the spaces created.

We were provided with layout plans of the building that Jim Mills had obtained from the previous owners which formed the basis of our layout proposal. Jim Mills is now claiming that our drawings show an overall floor area approximately 30% greater than in fact exists. ABC Holdings, who are part of this action, is a special purpose vehicle (SPV) that was set up sometime after he commissioned us to carry out the feasibility study.

The claimants state that they paid £116,000 more for the property than it was worth, due to the information we provided. We have emphasised that NDG undertook the commission in good faith, to provide a sketch proposal on the basis of the drawings provided to Jim Mills by the previous owner, which had come from a reputable firm of surveyors.

We have notified our Professional Indemnity Insurance providers of this claim and are awaiting the appointment of solicitors and confirmation of an initial meeting. Our response has been to state that:

- We did exercise care and skill in preparing the feasibility/concept drawings, setting out a potential layout using the information provide and without measurements of the property.
- We have denied any breach of duty of care to Jim Mills
- We also deny any duty of care to ABC Holdings who we did not have a contract with and who did not exist until sometime after the proposals were drafted.
- We affirm that our client has contributory negligence for not obtaining an accurate survey by instructing surveyors.

**QUESTION 7**  
**Page 2 of 2**

*EMAIL*

*CONTINUATION SHEET*

Date: 02 March 2021

From: Michael Hart

Re: Acorn Centre, Madchester: Re-development of the existing listed building

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- We also maintain that the client could have contacted the surveyor who produced the survey, that was in turn given to us - but Jim Mills probably did not as he did not want to incur any costs.

In the meantime, I would value your thoughts on the claims and their validity. I appreciate that you are still to qualify, but I know, from our mentoring conversations, that this is an area you have covered and enjoyed discussing in your Part 3 studies.

Many thanks

Belinda

## QUESTION 8

Page 1 of 3

*Sue.Allen@ndgarchitects.com*

EMAIL

Date: 02 March 2021

To: Candidate

From: Michael Hart

Re: **Acorn Centre, Madchester: Re-development of the existing listed building**

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Good morning,

As you are aware, the concept for the re-development of the Grade II building for the Acorn Centre was progressed at pace by the practice during the pandemic/latter part of 2020, as part of an initial feasibility study that was subsequently used to create interest, seek support and make applications for funding. Following confirmation that funds have been secured through the Local Authority, NDG has now been appointment under a RIBA Concise PSC 2020 for Conservation Architectural Services using traditional procurement and has been requested to set out the next steps to move the ambitious sketch proposals forward. An appointment has also been agreed for Principal Designer services under a separate PSC. The client is somewhat nervous regarding the extent of the proposed works to the listed building and various conditions attached to the principal source of funding.

You have worked with me before and yet again drawn the short straw, but we have the potential to make this project something quite special! The feasibility study has provided us with accurate site and building drawings, an assessment of condition and specialist surveys which have confirmed that no hazardous materials are present, foundations are set below the proposed reduced lower ground floor level and the envelope structure is sound, but we do seem to have the occasional visit from some pipistrelle bats and a number of pigeons are using the roof space as shelter. Whilst the severe dilapidation will necessitate major repair and refurbishment, the conditions imposed by the principal funding body to reinstate the Georgian details whilst accommodating twenty-first century needs will give us some interesting challenges. You will see from the attached photographs the extent of deterioration, but there are enough surviving details for us to restore the building to its former glory whilst integrating contemporary elements to provide new accommodation, services and facilities. I hope you are excited by the opportunity rather than depressed by what you see!

The concept proposals to be developed comprise:

- Full repair, restoration and refurbishment of the listed building
- Reduced lower ground floor to provide improved (and sensible) headroom. We are reducing the existing floor level by approximately 400mm

**QUESTION 8**  
**Page 2 of 3**

*EMAIL*

*CONTINUATION SHEET*

Date: 02 March 2021

From: Michael Hart

Re: **Acorn Centre, Madchester: Re-development of the existing listed building**

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- Rear extension to accommodate level access/lift installation to the upper floors/new toilet and support accommodation
- Café located at the lower ground floor with integrated display/interpretation about the site and building
- Flexible teaching and lettable incubation spaces at ground, first and second floor levels
- Community garden and adjustment of external levels within the site boundary
- External and internal signage

The previous use of the building was residential but there is established educational and administrative use in the abutting terraced accommodation to which we will be adding the additional spaces and facilities.

There is a lot to do, but in order for us to be well prepared for the forthcoming Project Team meeting, can I ask you please to think about the following items and send me a memo outlining the relevant points and details. Keep the details specific to the project. You do not need to impress me with your wider knowledge. You can do that when we chat later!!

1. What statutory applications will be required?
2. What specific information might we need to produce to explain and justify the proposals for what will be perceived as radical intervention to the listed building?
3. Do we need to carry out any further surveys or investigations?
4. Who else needs to be formally appointed to the Design Team for the development and implementation stages of the project?

Please contact me if you have any queries.

Thanks and regards

Michael

Michael Hart  
Director



**QUESTION 8**  
**Page 3 of 3**

**Existing Site and Building Information**



## QUESTION 9

Page 1 of 5

**NDG Architects**

### INTEROFFICE MEMO

Date: 15 December 2021 (Note date projected forward to Practical Completion)  
To: Candidate  
From: Sue Allen  
Re: **New House for Alec Smart**

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The Smart House project has now reached Practical Completion, thank goodness, and as you now know domestic clients can be very demanding! However, I have just looked at the office staff resources and got a bit of a shock. It seems we have spent almost twice as many person days as we should have done. I feel embarrassed because we have all been so busy, I had not noticed and therefore have not said anything to the client about this. I mentioned the situation to Alec Smart last week and he was clear he has no more money! He also pointed out that our percentage fee covered the increase in scope associated with the larger footprint. Nevertheless, I feel we now have no option but to put a case to the client for the additional fees.

We were appointed using the RIBA Standard Professional Services Contract 2020 and I attach a schedule listing the key aspects of the agreement. As you can see, we stated that our fee would be based on 7% of the agreed final contract sum. The problem is that Alec Smart has taken various elements out of the job to save money and thereby has reduced the scope of the main contract works for which we are acting as Contract Administrator. Whilst it is true that the client organised these contractors himself, he used the drawings we had produced and of course there was a lot of co-ordination needed by NDG as a result of his constant phone calls! The QS estimates the value of all work is around £700,000. However, the building contract we administered only has a contract sum of £500,000, and this is the figure upon which our percentage fee is based!

Alec Smart has not shared with us the value of the works he took out of the project and dealt himself, with, but the QS suggests they consist of £100,000 for the retaining wall, £50,000 for the bathrooms and £50,000 for the kitchen. Please see the attached schedule of fees which lists our original assumptions to arrive at a fee of 7% together with a schedule of the actual person days spent at each of the work stages. According to our records, we have spent 70 person days more than we targeted. Even at our break-even point of £400 per day this represents a loss of £21,000.

(For simplicity in the chart, I have provided an average day rate to cover all the people who worked on the project).

**QUESTION 9**  
**Page 2 of 5**

**MEMO**

**CONTINUATION SHEET**

Date: 15 December 2021

Re: **New House for Alec Smart**

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The key areas of additional work for NDG are as follows:

- After we had agreed a layout at RIBA Work Stage 3, the client decided to add 70 sq m of additional usable floor area by making the house 5 m wider. It was this scheme we then submitted for planning approval, which delayed the submission by 3 weeks and took us 15 person days to deal with the amendments (and increased the cost of the project by around £100,000).
- Taking the retaining wall out of the main building contract did not actually save us any professional time, in fact we spent 5 person days extra just dealing with the coordination of the design with the client's own contractor!
- The client also chose to organise the kitchen and bathroom fit-out outside the main construction contract but again, this still took 15 person days to co-ordinate the activities of the client's own specialist contractors. Also, this was design work we had already completed at RIBA Work Stage 4 including full specification of all the fittings!
- We planned for 6 meetings with the client but have minutes of 16 client meetings.
- The main contract we administered lasted for 15 months, 12 weeks longer than expected and we spent 5 person days dealing with the contractors claim at the end of the contract.

Could you please do the following

1. Draft a letter to the client setting out a detailed case for paying NDG additional fees including the figures.
2. Write a memo setting out lessons learned for the office in the future and any professional issues associated with the situation in which we find ourselves.
3. Finally, would the project have been profitable if the retaining wall, kitchen and bathroom works had been included within the main contract?

Thanks and best regards

Sue



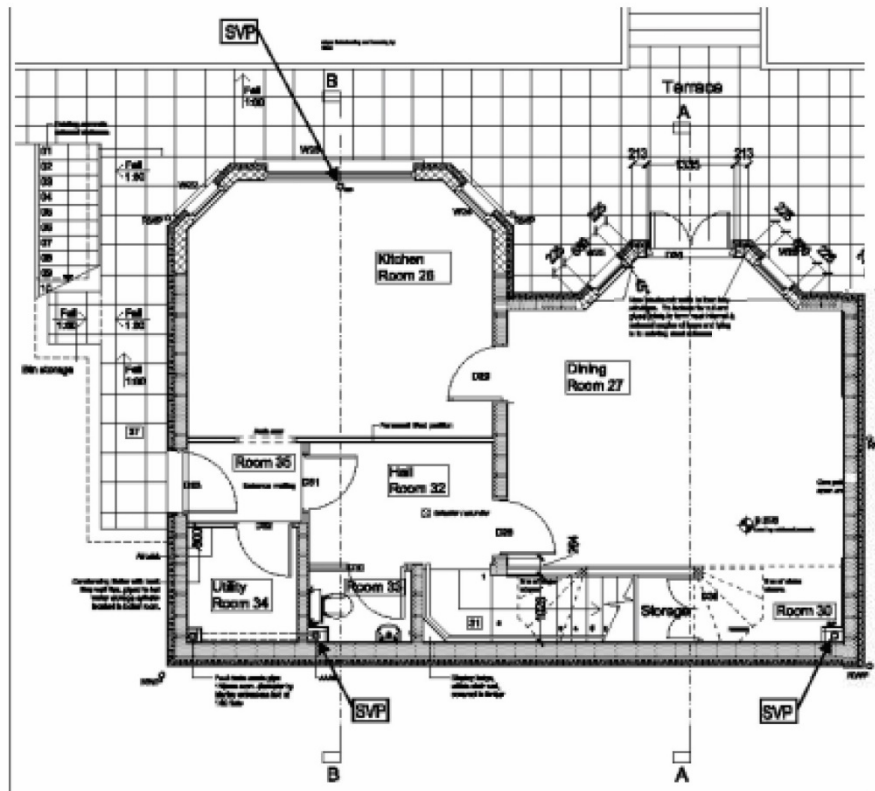
**QUESTION 9**  
**Page 3 of 5**

**NEW HOUSE for ALEC SMART**  
**SCHEDULE OF FEES and SERVICES**

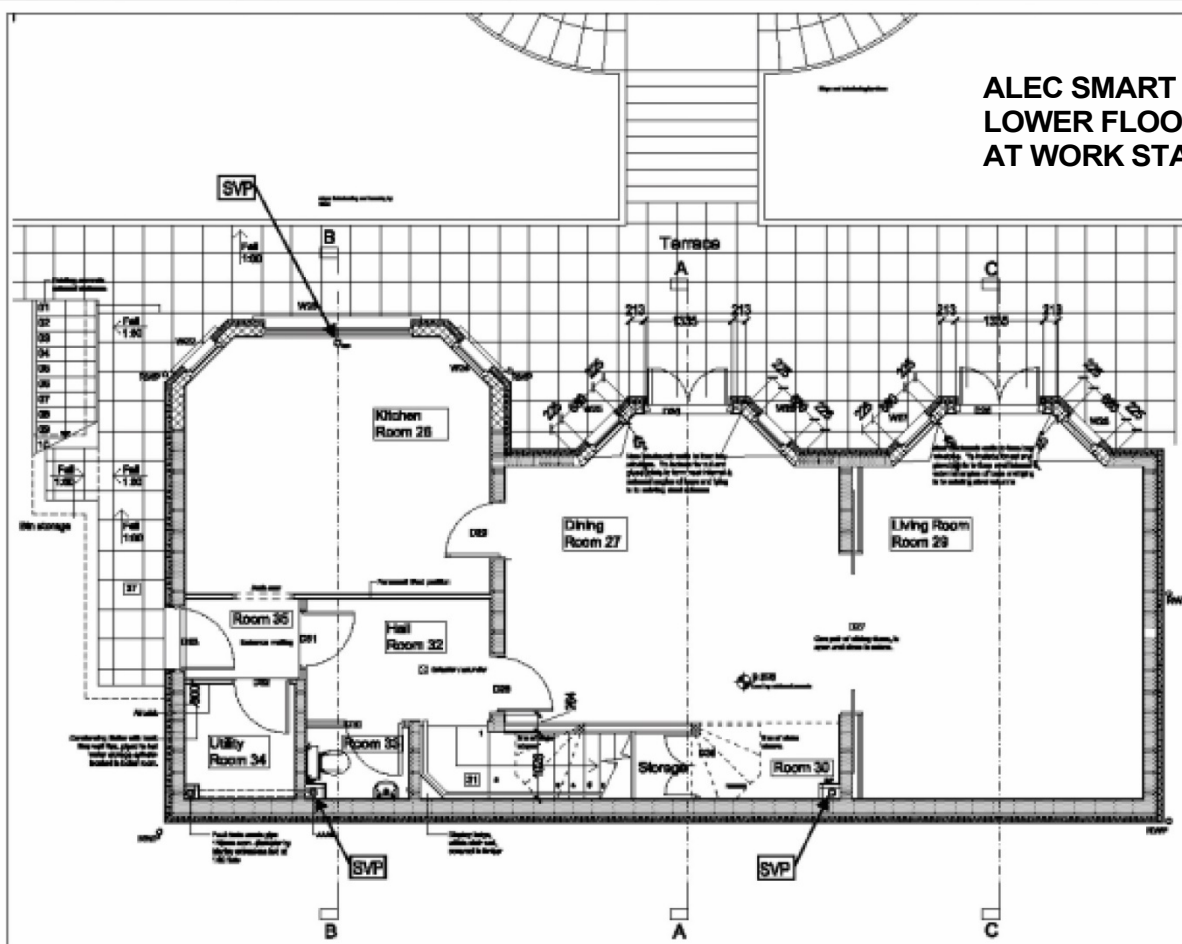
|                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Items A+B - client and consultant names                                                             | Client – Mr Alec Smart<br>Architect - NDG                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Item C site address                                                                                 | Plot 4, The Precipice, Madchester M22 1EE                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Item D Project brief – construction cost, programme,                                                | Brief – design of a single dwelling on a sloping site<br>Construction budget £500,00<br>Programme<br>RIBA stages 1,2,3 – 4 months<br>RIBA stages 4 – 3 months<br>RIBA stages 5,6 – 12 months<br>RIBA stage 7 – 12 months                                                                                                                                                                                                                                          |
| Item E Other consultants                                                                            | QS – PDQ associates<br>Structural Engineer – AP Strong<br>M&E Engineer – Planet Ltd<br>Principal Designer - NDG                                                                                                                                                                                                                                                                                                                                                   |
| Items FGHI – basic fee calculation, no of meetings, rates for time charge, Expenses, Invoice points | Fee basis 7% of the final agreed contract sum<br><br>Fee Instalments<br>10% of fee at completion of stage 1+2<br>20% of fee at completion of stage 3<br>30% of fee at completion of stage 4<br>15% of fee at 50% completion of stage 5<br>15% of fee at 100% completion of stage 5,6<br>10% of fee at completion of 12 months defects<br><br>6 meeting with Client<br>Expenses included within above figures<br>Charge out rates as attached NDG schedule of fees |
| Item j – PI Insurance                                                                               | £1m cover                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Item K dispute resolution                                                                           | Adjudication                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| L – information formats                                                                             | 2D AutoCAD and PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| M – supplementary rights                                                                            | None                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Schedule of services – Role specialist services, additional and other services                      | NDG – Architect, Lead Designer, Project Manager, Contact Administrator (traditional contract)<br>RIBA Work stage 1 - 7                                                                                                                                                                                                                                                                                                                                            |
| Contract conditions                                                                                 | As Standard PSC 2020                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**QUESTION 9**  
Page 4 of 5

**ALEC SMART HOUSE  
LOWER FLOOR PLAN  
AT WORK STAGE 3**



**ALEC SMART HOUSE  
LOWER FLOOR PLAN  
AT WORK STAGE 4**



**QUESTION 9****Page 5 of 5****NEW HOUSE for ALEC SMART  
SCHEDULE OF PLANNED AND ACTUAL FEES**

| RIBA WORK STAGE                                                  | PERSON DAYS PROPOSED               | PERSON DAYS ACTUALLY SPENT |
|------------------------------------------------------------------|------------------------------------|----------------------------|
| Preparation and brief (1)                                        | 5                                  | 10                         |
| Concept and developed design (2+3)                               | 15                                 | 30                         |
| Tender and technical (4)                                         | 20                                 | 35                         |
| Construction (5+6)                                               | 25                                 | 65                         |
| Defects and aftercare (7)                                        | 5                                  | Nothing yet!               |
| <b>Total Person Days</b>                                         | <b>70</b>                          | <b>140</b>                 |
| Total cost to office - (Break even point of £400 / person / day) | 28,000                             | 56,000                     |
| Add Profit @ 25%                                                 | 7,000                              | NIL                        |
| Total fee                                                        | <b>£35,000</b><br>(7% of £500,000) |                            |

## QUESTION 10

Page 1 of 2

*Belinda.Carter@ndgarchitects.com*

EMAIL

Date: 04 March 2021

To: Candidate

From: Belinda Carter

Re: **Tappers Dance Studio: Contractor in Administration**

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Hi,

We are all working from home and it is slowing down the receipt of post via the office. Consequently, I have only just received the attached letter. Time is short (cynically, I think this may have been timed deliberately), so I would appreciate your thoughts putting together:

1. A letter in response to the letter received from the solicitor (below)
2. Some thoughts on what advice we need to give to the client.

We can then review it via a '*Teams*' meeting, with a view to issuing it tomorrow:

You will remember the project for Tappers Dance Studio. It reached Practical Completion 11 months ago, and the client was delighted with the result, the way we ran the project, and the overall relationship of all parties involved. It all went relatively smoothly. There were a few moments along the way, but everyone seemed to have come out of it well. Unfortunately, I have just heard that the contractor Jolie Construction went into administration a number of weeks ago, which is sad because they seemed genuinely a good company and did an excellent job. We are lucky that it was not during the construction period, as it could have been far worse for the project, but even so, we have received the attached letter and need to act.

Can you draft a formal letter in response? Given that it is from a solicitor and appears rather legal, please make sure we respond in a very professional manner. If you look at the dates, I believe we may need to move quickly, so I want you to put this together as fully as you can. Then, we must advise the client about what needs to be done and what are our duties as the Contract Administrator and what we can do as the Project Architect and Principal Designer (I do not think we need to list everything, just pick up what you think may be the key issues so we can talk about them).

Many thanks.....Belinda

**QUESTION 10**  
**Page 2 of 2**

**Copy of the Letter from Solicitors**

Strongarm, Foot and Bowels Solicitors LLP  
1st Prime Suite  
1 Queen St  
Madchester

FAO The Contract Administrator for the Tappers Dance Studio  
**NDG Architects**  
Old Mill Building  
Madchester  
MX1 9DS

1st March 2021

Dear Sir or Madam,

**Re: JOLIE CONSTRUCTION LIMITED - IN ADMINISTRATION**  
**TAPPERS DANCE STUDIO**

We act on behalf of Baking Thilly Restructuring and Recovery LLP, the Joint Administrators of the above Company in relation to settlement of outstanding final accounts and other contractual matters.

A copy of the Joint Administrators' Appointment is available upon request, together with a copy of our instructions in relation to this matter. It would be appreciated if all further correspondence could be sent to ourselves and not to the Company's former offices.

We note from the Company's records that a Final Account for the above contract has been agreed with a sum of £32,000.00 (excluding VAT) presently outstanding, in respect of retention monies relating to the above contract. We shall therefore be pleased to receive your Final Certificate releasing the retention, together with a cheque in settlement.

Should the retention monies not yet be due for release at this time, then it would be appreciated if you could confirm the anticipated date for the issue of the Making Good of Defects Certificate and the Final Certificate, following which we presume that payment of retention can then be anticipated. Please note that as the Company is subject to Insolvency proceedings, all final documentation together with your cheque should be issued to Jolie Construction Ltd - In Administration" and be forwarded to our address as detailed above.

Alternatively, if payment is to be made via BACS transfer, please remit to the following account:

Account Name: Jolie Construction Ltd - In Administration  
Account No:00123456      Sort Code:66-66-66

We understand that the Contract is a JCT Standard Building Contract 2016 with Quantities, and you are the Contract Administrator. We understand that you are aware of your duties under the contract and given that the Rectification Period is due to conclude on 14<sup>th</sup> March 2021 we require your response within 7 days of this letter. If you fail to do so, the full amount will be expected to be paid and a Final Completion Certificate issued by or on the 14<sup>th</sup> March 2021 instructing the client to pay the full amount.

Yours faithfully

A Toole Esq.

**END OF DAY TWO**

**Please ensure that you email a copy of your draft answers  
to your exam supervisor  
at 1730hrs**

**&**

**Please ensure that you upload a copy of your draft answers  
to the box.com link (provided in the exam scenario email sent 24 February 2021)  
at 1730hrs**